

Map Template (County) — Reference Document

Animated, Interactive, and Choropleth Network Visualization for U.S. Counties

2026-06-26

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1 Overview

map_template_county.r visualizes pairwise link data (migration flows, trade, firm networks, etc.) across U.S. counties over time. Nodes are counties, keyed by their zero-padded five-digit FIPS code

(`GEO_ID`). It produces:

1. **Heatmap animation** — MP4/GIF coloring counties by their number of connections (degree) or summed weight (strength), with in/out/all modes.
2. **Network animation** — MP4/GIF drawing filtered edges on a county base map, year by year.
3. **Node-value choropleth animation** — MP4/GIF mapping any per-county scalar (total inflow, outflow, or net migration) on a sequential or diverging color scale.
4. **Interactive map** — zoomable HTML leaflet with a year dropdown.
5. **Combined animation** — network edges overlaid on the degree heatmap.

All visualizations use **fixed global scales** across frames so year-to-year comparisons are valid. Counties that fail to match the base shapefile (Alaska, Hawaii, territories, and renamed/created codes) are dropped, not fixed. **Pass a single year to any `animate_*` function and it writes a single PNG instead of a video** — the basis for static, publication-ready snapshots.

2 Input Data Format

Edgelist — a `data.table` with these columns:

Column	Type	Description
<code>from</code>	character	Origin county FIPS (5-digit <code>GEO_ID</code>)
<code>to</code>	character	Destination county FIPS
<code>weight</code>	numeric	Edge weight (e.g., number of migrants)
<code>year</code>	integer	Year of observation

Note: `animate_heatmap()` does not require `weight`.

Node-value table — for `animate_county_choropleth()`, a `data.table` with columns `GEO_ID` (5-digit FIPS), `year` (integer), and `value` (numeric). `node_flows()` builds one of these directly from an edgelist.

3 Global Constants

Set these once at the top of the file (Section 1, `USER PARAMETERS`). Functions use them as defaults.

Constant	Default	What it controls
COUNTY_SDF_PATH	<i>path</i>	Pre-built county shapefile (<i>.rds</i>)
STATE_SDF_PATH	<i>path</i>	Pre-built state overlay (<i>.rds</i>)
OUTPUT_DIR	<i>.../output_county</i>	Where figures are written
PLOT_CAPTION	<i>""</i>	Default plot caption
TOP_PCT	0	Quantile threshold for "threshold" filter; 0 keeps all—raise (e.g. 0.999) for dense county maps
EDGE_COLOR	"#0072B2"	Default edge color
WIDTH_RANGE	<i>c(0, 1.5)</i>	Min/max edge line width (thin for dense maps)
ALPHA_RANGE	<i>c(0, 0.6)</i>	Min/max edge transparency
LOG_EDGES	FALSE	Log-transform edge width/alpha?
BIN_EDGES	0	Round weights to this unit (0 = continuous)
FILL_LOW	"white"	Low end of choropleth color ramp
FILL_HIGH	"#D55E00"	High end of choropleth color ramp

4 Edge Filtering Methods

The network functions accept a `filter_method` parameter:

4.1 **"threshold"** (default)

Keep edges above a global weight percentile.

- `top_pct` — quantile cutoff. Example: 0.999 keeps the top 0.1% of edges by weight. (County networks are dense, so a low cutoff produces a hairball; use a strict value.)

4.2 **"topk"**

Keep each node's K strongest connections, so every connected node is represented regardless of absolute weight.

- `topk_K` — top edges to keep per node (default 3).
- `topk_min_weight` — absolute floor below which edges are dropped even if top- K (default 0).

For undirected networks, edges are ranked from both endpoints and kept if top- K for either side; for directed networks, only outgoing edges are ranked per source.

4.3 **"disparity"** (backbone extraction)

Implements the Serrano et al. (2009) disparity filter. Keeps edges that are statistically significant relative to their endpoint's total traffic, at every scale simultaneously.

- `disparity_alpha` — significance level (default 0.05); lower values give a sparser backbone.
- `topk_min_weight` — absolute floor (default 0).

An edge is kept if its share of a node's total weight is surprisingly large:

$$\alpha_{ij} = (1 - p_{ij})^{k_i - 1},$$

where $p_{ij} = w_{ij}/s_i$ is the edge's share of node i 's total strength and k_i is node i 's degree. The edge is kept if α_{ij} falls below the threshold from **either** endpoint.

4.4 "absolute"

Keep edges whose raw weight clears a fixed floor.

- `abs_threshold` — keep edges with `weight >= abs_threshold` (default 0). Useful as a pass-through when feeding an edgelist that is already filtered (e.g. a pre-computed backbone) while still letting the function fix one shared scale across years.

5 Functions

5.1 `animate_heatmap()` — Degree/Strength Choropleth Animation

Each frame colors counties by how connected they are that year.

Parameter	Default	Description
<code>edgelist</code>	<i>required</i>	<code>data.table</code> (from, to, year)
<code>snap_years</code>	all years	Integer vector of years to animate
<code>directed</code>	FALSE	Treat edges as directed?
<code>degree_mode</code>	"all"	"all", "in", or "out" connections
<code>node_metric</code>	"degree"	"degree" (count of partners) or "strength" (sum of weights; needs <code>weight</code>)
<code>fill_low</code>	FILL_LOW	Low-end color
<code>fill_high</code>	FILL_HIGH	High-end color
<code>title_prefix</code>	"Network Connectivity"	Title before year label
<code>subtitle</code>	degree description	Plot subtitle
<code>caption</code>	PLOT_CAPTION	Plot caption
<code>fps</code>	1	Frames per second
<code>fixed_max</code>	NULL	Override color-scale max so several heatmaps share one scale
<code>save_path</code>	.../animation_heatmap.mp4	Output file (.mp4/.gif, or .png if one year)

```
# In-migration connectivity, pinned scale so in/out maps match
animate_heatmap(mig_county, snap_years = 1990:2021,
  directed = TRUE, degree_mode = "in",
  fill_high = "#009E73", fixed_max = 792,
  save_path = "output/indegree.mp4")
```

5.2 `animate_network()` — Network Edge Animation

Each frame draws filtered edges on a county base map; width and opacity encode weight.

Parameter	Default	Description
<code>edgelist</code>	<i>required</i>	<code>data.table</code> (from, to, weight, year)
<code>snap_years</code>	all years	Years to animate
<code>scale_years</code>	NULL	Years used to fix the global edge scale (defaults to <code>snap_years</code>); widen it to share one scale across separate single-year calls
<code>directed</code>	FALSE	If TRUE, draw arrows
<code>filter_method</code>	"threshold"	"threshold", "topk", "disparity", "absolute"
<code>top_pct</code>	TOP_PCT	Quantile cutoff (threshold only)
<code>topk_K</code>	3	Edges per node (topk only)
<code>topk_min_weight</code>	0	Absolute floor (topk & disparity)
<code>disparity_alpha</code>	0.05	Significance level (disparity only)
<code>abs_threshold</code>	0	Weight floor (absolute only)
<code>edge_color</code>	EDGE_COLOR	Edge color
<code>log_edges</code>	LOG_EDGES	Log-transform widths?
<code>bin_edges</code>	BIN_EDGES	Weight binning unit (0 = continuous)
<code>title_prefix</code>	"Network Connections"	Title before year label
<code>caption</code>	PLOT_CAPTION	Plot caption
<code>fps</code>	1	Frames per second
<code>save_path</code>	<code>.../animation_network.gif</code>	Output file (.mp4/.gif, or .png if one year)

```
# Backbone snapshots that share one edge-width scale across years.
# scale_years spans all the years you will render separately, so a
# corridor of a given weight draws identically in each PNG.
for (yr in c(1990, 2000, 2010, 2021)) {
  animate_network(mig_county, snap_years = yr,
    scale_years = c(1990, 2000, 2010, 2021),
    directed = TRUE, filter_method = "disparity",
    disparity_alpha = 0.10, log_edges = TRUE,
    edge_color = "#009E73",
    save_path = sprintf("output/net_%d.png", yr))
}
```

5.3 `animate_county_choropleth()` — Node-Value Choropleth Animation

Maps any per-county, per-year scalar (total inflow/outflow, net migration, or any `GEO_ID`-year measure). Unlike `animate_heatmap()`, which derives a value from network degree, this takes values directly, so it supports signed quantities via a diverging scale centered at zero. Limits are fixed across frames.

Parameter	Default	Description
<code>node_dt</code>	<i>required</i>	<code>data.table</code> (GEO_ID, year, value)
<code>snap_years</code>	all years	Years to include
<code>diverging</code>	FALSE	TRUE → scale centered at 0 (for signed values like net migration)
<code>fill_low</code>	FILL_LOW	Low / most-negative color
<code>fill_mid</code>	"white"	Midpoint color (diverging only)
<code>fill_high</code>	FILL_HIGH	High / most-positive color
<code>legend_name</code>	"Value"	Legend title
<code>title_prefix</code>	"County Choropleth"	Title before year label
<code>subtitle</code>	""	Plot subtitle
<code>caption</code>	PLOT_CAPTION	Plot caption
<code>fps</code>	1	Frames per second
<code>fixed_max</code>	NULL	Color-scale max (abs value if diverging); values beyond are squished to the extreme color, not dropped
<code>cap_pct</code>	NULL	Quantile in (0, 1] to cap the scale (of value when diverging); tames outlier metros. Ignored if <code>fixed_max</code> is set
<code>save_path</code>	.../animation_choropleth.mp4	Output file (.mp4/.gif, or .png if one year)

```

nf <- node_flows(mig_county)          # per-county inflow / outflow / net

# Total inflow (sequential), capped at the 98th percentile
animate_county_choropleth(nf[, .(GEO_ID, year, value = inflow)],
  snap_years = 1990:2021, cap_pct = 0.98, fill_high = "#009E73",
  legend_name = "In-migrants", save_path = "output/inflow.mp4")

# Net migration (diverging): blue = net gain, red = net loss
animate_county_choropleth(nf[, .(GEO_ID, year, value = net)],
  snap_years = 1990:2021, diverging = TRUE, cap_pct = 0.98,
  fill_high = "#2166AC", fill_mid = "white", fill_low = "#B2182B",
  legend_name = "Net migration", save_path = "output/net.mp4")

```

5.4 `interactive_network()` — Interactive Leaflet HTML Map

A zoomable, clickable map. One year gives a single map; multiple years add a year dropdown.

Parameter	Default	Description
<code>edgelist</code>	<i>required</i>	<code>data.table</code> (from, to, weight, year)
<code>snap_years</code>	<i>required</i>	Integer vector (1 = single map, 2+ = dropdown)
<code>directed</code>	FALSE	If TRUE, add circle markers at destinations
<code>filter_method</code>	"threshold"	"threshold", "topk", "disparity"
<code>top_pct</code>	TOP_PCT	Quantile cutoff
<code>topk_K</code>	3	Edges per node
<code>topk_min_weight</code>	0	Absolute floor
<code>disparity_alpha</code>	0.05	Significance level
<code>edge_color</code>	EDGE_COLOR	Edge color
<code>fill_low</code>	FILL_LOW	Low-end choropleth color
<code>fill_high</code>	FILL_HIGH	High-end choropleth color
<code>title</code>	"Network"	Map title
<code>save_path</code>	.../interactive_network.html	Output HTML file

```
interactive_network(mig_county,
  snap_years = c(1995, 2000, 2005, 2010, 2015),
  filter_method = "disparity", edge_color = "#009E73",
  fill_high = "#009E73", save_path = "output/interactive.html")
```

5.5 `animate_combined()` — Heatmap + Network Overlay Animation

Each frame shows the degree choropleth with filtered edges drawn on top.

Parameter	Default	Description
<code>edgelist</code>	<i>required</i>	<code>data.table</code> (from, to, weight, year)
<code>snap_years</code>	all years	Years to animate
<code>directed</code>	FALSE	If TRUE, draw arrows
<code>filter_method</code>	"threshold"	"threshold", "topk", "disparity"
<code>top_pct</code>	TOP_PCT	Quantile cutoff
<code>topk_K</code>	3	Edges per node
<code>topk_min_weight</code>	0	Absolute floor
<code>disparity_alpha</code>	0.05	Significance level
<code>edge_color</code>	EDGE_COLOR	Edge color
<code>log_edges</code>	LOG_EDGES	Log-transform widths?
<code>bin_edges</code>	BIN_EDGES	Weight binning unit
<code>fill_low</code>	FILL_LOW	Low-end choropleth color
<code>fill_high</code>	FILL_HIGH	High-end choropleth color
<code>title_prefix</code>	"Network Connections"	Title before year label
<code>subtitle</code>	overlay description	Plot subtitle
<code>caption</code>	PLOT_CAPTION	Plot caption
<code>fps</code>	1	Frames per second
<code>save_path</code>	.../animation_combined.mp4	Output file (.mp4 or .gif)

6 Helper Functions

These are called internally by the main functions but can also be used standalone.

6.1 `node_flows(el)`

Collapses a directed weighted edgelist into per-county flow totals. Returns a `data.table` with columns `GEO_ID`, `year`, `inflow` (sum of incoming weights), `outflow` (sum of outgoing weights), and `net` (`inflow - outflow`; positive = net receiver). Counties present in only one direction get 0 on the missing side.

```
nf <- node_flows(my_edges) # my_edges: from, to, weight, year
```

6.2 `filter_topk(yr_edges, K, min_weight, directed)`

Returns a filtered `data.table` of each node's K strongest edges.

```
filtered <- filter_topk(my_edges, K = 3, min_weight = 100)
```

6.3 `filter_disparity(yr_edges, alpha, min_weight, directed)`

Returns a filtered `data.table` of backbone edges (Serrano et al. 2009).

```
filtered <- filter_disparity(my_edges, alpha = 0.05)
```

6.4 `save_frames(frame_paths, save_path, fps)`

Stitches frames into an `.mp4/.gif` by the `save_path` extension. **If only one frame is supplied it writes a single `.png` instead**, rewriting the extension — this is why passing a single year to any `animate_*` function yields a static image.

7 Quick-Start Recipe

```
# 1. Source the template (edit Section 1 paths first)
source("map_template_county.r")

# 2. Load your edgelist -- columns: from, to, weight, year
#    (from/to are 5-digit county FIPS strings)
my_data <- fread("my_county_edgelist.csv")

# 3a. A network animation (backbone), one shared scale across years
animate_network(my_data, snap_years = 2000:2020,
                filter_method = "disparity", disparity_alpha = 0.10,
                save_path = "output/my_network.mp4")

# 3b. A net-migration choropleth from the same edgelist
nf <- node_flows(my_data)
animate_county_choropleth(nf[, .(GEO_ID, year, value = net)],
                          snap_years = 2000:2020, diverging = TRUE, cap_pct = 0.98,
                          legend_name = "Net migration", save_path = "output/net.mp4")
```


8 Output Formats

- `.mp4` — Video (requires the `av` package). Frames are duplicated internally for smooth playback.
- `.gif` — Animated GIF (requires `magick`). Larger, but universally viewable.
- `.png` — Single static frame, produced automatically whenever a function is given exactly one year.
- `.html` — Interactive leaflet map. Opens in any browser; supports zoom, pan, hover tooltips, and year selection.